

# Recent Kolmogorov–Arnold Network (KAN) Architecture -level developments

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## Purpose and Scope

**Note : no paper using KAN in single-molecule localization microscopy, or PSF-based emitter localization was found in this window. This is consistent with the KAN-SMLM direction remaining a novel, unclaimed intersection as of this search.**

## 1. Optimized Architectures for Kolmogorov–Arnold Networks

*Bagrow, J. & Bongard, J.*

Citation: James Bagrow and Josh Bongard, "Optimized Architectures for Kolmogorov–Arnold Networks," Apr 20, 2026.

Link: <https://arxiv.org/abs/2512.12448>

### What the paper does

The paper addresses a stated tension: architectural enhancements that improve KAN accuracy (skip connections, added depth, DenseNet-style blocks) tend to reduce the interpretability that motivates using KANs in the first place. The authors propose training an overprovisioned KAN together with a differentiable sparsification and depth-selection mechanism, so that a compact, interpretable model is discovered during training rather than chosen by hand.

Three architectural mechanisms are combined:

- Edge gates ("egates") — a differentiable relaxation of  $\ell_0$  regularization (following Louizos et al., 2018) applied per activation-function edge, allowing individual learnable spline activations to be pruned to exactly zero during training.
- DenseNet-style forward connections ("FCs") — each layer receives the concatenation of all previous layer outputs (and the original input), providing deep supervision and letting the network skip unnecessary depth.
- Multi-exit / exit gates ("xgates") — every layer gets its own output head; a differentiable categorical (Gumbel-Softmax) gate learns which exit to use as the final output, providing explicit depth selection.

Training is guided by a minimum description length (MDL) objective that jointly penalizes prediction error and the number of "open" gates (edges, nodes, and the active exit), following a BIC-style approximation for model complexity.

### Reported results

The authors ran a  $2 \times 2 \times 2$  factorial ablation (egates present/absent  $\times$  forward connections present/absent  $\times$  exit gates present/absent) across three problem classes: symbolic function-approximation benchmarks (the first 10 Nguyen problems), two dynamical-systems forecasting tasks (the Ikeda map and a 3-species ecosystem model), and two real-world regression datasets (concrete compressive strength, superconductor critical temperature).

- Edge gates alone ("E") were reported as necessary but not sufficient: condition E reduced model size but consistently harmed accuracy relative to baseline across the Nguyen problems ("E led to worse accuracy than baseline on all problems").
- Conditions combining edge gates with a depth-selection mechanism — EF, EX, or EFX — were reported as the best performers, generally producing models that were both smaller and more accurate than the standard KAN baseline.

- On the Ikeda map, the EX condition matched baseline accuracy using 16 contributing edges versus 48 for baseline, per the paper's supplementary tables.
- On the concrete-strength dataset, the EFX condition at one regularization setting ( $\beta = 0.01$ ) was reported to achieve 4.87 MPa RMSE versus the baseline's 4.91 MPa, using a model with 64 contributing edges versus 351 for baseline (about 18% of the size).
- A Pareto hypervolume analysis (trading off RMSE, number of contributing edges, and compositional depth) ranked condition EX highest on average across the five problem domains, with EF and EFX also ranking well; no single condition dominated on every domain.

The authors also list an explicit limitation: their differentiable gating mechanism is noted to have known variance issues during training, and the ecosystem dynamical-systems task "was prone to overregularization," indicating the method is not push-button and needs hyperparameter tuning of the sparsity weight  $\beta$  and initialization values.

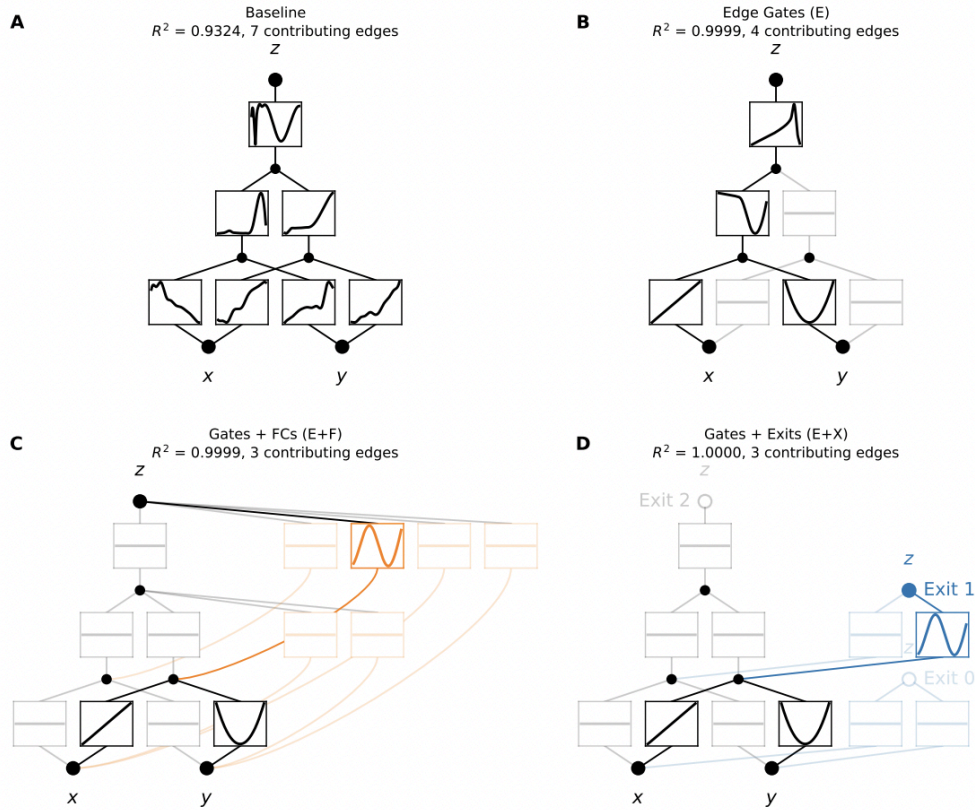


Figure 1: Learning  $z = \sin(x + y^2)$  with overprovisioned KANs (architecture [2, 2, 1, 1]). Forward connections are shown in orange and exit gates in blue.

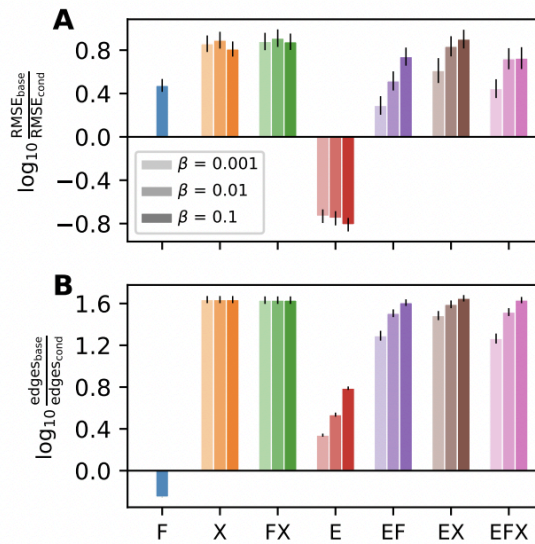


Figure 2: Model accuracy and size relative to KAN baseline. Log-ratios of (A) test RMSE and (B) contributing edges, pooled across Nguyen problems F1–F10 (see also App. B) and broken down by condition and  $\beta$ . Positive values indicate conditions have lower RMSE or fewer contributing edges than baseline. Errorbars denote  $\pm 1$  SEM on log-ratios.

## 2. Clifford Kolmogorov-Arnold Networks (CIKAN)

Wolff, M., Alesiani, F., Duhme, C., Jiang, X.

Citation: Matthias Wolff, Francesco Alesiani, Christof Duhme, Xiaoyi Jiang, "Clifford Kolmogorov-Arnold Networks,"

Link: <https://arxiv.org/abs/2602.05977>

## 3. Kolmogorov-Arnold Fourier Networks

Zhang, J. et al. — Abstract/snippet only

Citation: Jusheng Zhang (Sheng) et al., "Kolmogorov-Arnold Fourier Networks," arXiv:2502.06018. Originally submitted Feb 9, 2025; v3 revision posted May 24, 2026 (the revision falling in the window of interest).

Link: <https://arxiv.org/abs/2502.06018>

Per the retrieved snippet, a key contribution stated is "a fundamental basis transformation from the local, grid-based B-spline representation to a global, adaptive spectral representation" — i.e., replacing the standard B-spline parameterization of the edge activation functions with a Fourier/spectral basis. Only the abstract-level snippet was available; the full methodology and quantitative results in the May 2026 revision were not retrieved in this session and should be checked directly before citing specifics.

- RSS-KAN (Zhang, Fan & Lu, 2026), published in Neurocomputing (DOI 10.1016/j.neucom.2026.132796): per a citing paper's one-line description, proposes replacing B-spline activations with ReLU, sigmoid, and sine bases. Paywalled; not independently verified from the primary source.
- Faroughi, Mostajeran, Mashhadzadeh & Faroughi, "Kolmogorov-Arnold networks for data-driven, physics-informed, and deep-operator learning: a review, synthesis, and new analysis," Neural Networks, DOI 10.1016/j.neunet.2026.108791 (epub March 1, 2026). This is a review/survey rather than a new architecture, covering the shift from MLPs to KANs in scientific machine learning. Paywalled; abstract only.